When a Local Pricing Guard Is Not a Welfare Promise

Finite precision, continuing intervention and adaptive behavior in a bounded logit market

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Abstract. A local pricing rule can be valid on every execution without supporting the customer-benefit promise attributed to it. We study a symmetric, bounded logit duopoly whose externally enforced guard permits prices with near-best immediate profit against the rival's previous public price. In a coarse inherited game, one real price-cap round can move all later permitted behavior into a favorable region. That apparent protection weakens under richer price choices: the base continuous worst long-run customer benefit is about 66.272, below the unchanged relaxed 95% target of 68.513. A tight upper-price envelope then characterizes a repair within one declared family: repeat the same physical cap every K rounds. The base target permits K=4, while K=5 fails. We separately quantify a bounded demand-calibration reserve and small unilateral profit improvements against the adverse public script; a feasible worst path is not automatically a selfish equilibrium. Finally, 192 prospectively specified local tabular-learning runs compare four cap schedules, two initializations and twelve paired market seeds under two masks in one calibration. The four paired mean customer-benefit changes range from +2.391 to +2.946, conditional on the fixed design. Saved-policy graph audits separate actual training, frozen continuation and all-policy protection. The contribution is a reproducible, model-specific guarantee audit and graphical benchmark, not a new reinforcement-learning algorithm, a general anti-collusion solution or evidence about language-model sellers.

1 The problem: a local check and a long-run promise

Two shops choose prices simultaneously. Each wants its own profit. An external guard checks a proposed price against the other shop's *last completed* price. Passing that check says little by itself about what customers obtain after many executions. We ask what customer-benefit promise survives a one-time handover back to this guard, and what continuing control is needed for a stronger promise.

A cap is a real market round: it limits both executed prices, makes purchases and earns profit and customer surplus. It is not a free reset of a chart. Our human game deliberately lets a coordinator choose both shops' prices. That illustrates permitted behavior; it does not establish independent collusion or supply human experimental data.

What is added. The linked sprint starter already provides replacement, history, objective and checkpoint-branch experiments [1]. We instead audit the capacity of an explicit external price rule, including finite-to-continuous transfer, an attainable worst envelope, same-cap timing, model reserve and limited local learning. Known control and learning methods are credited below; priority for these ingredients is not claimed.

2 A self-contained game and the scope of each claim

There are B potential buyers, two symmetric inside goods of quality v and cost c, an outside option of normalized utility zero, and taste scale mu>0. At executed physical prices p_A,p_B, let e_i=exp((v-p_i)/mu). Expected demand, private profit and customer surplus are:

```
D_i = B*e_i / (1+e_A+e_B)
R_i = (p_i-c)*D_i
C(p_A,p_B) = B*mu*log(1+e_A+e_B)
```

The base calibration is B=100, v=2, c=1 and mu=0.25. Its inherited bounded physical price domain is approximately [1.427721, 1.970186]. The inherited cap is held exactly at its saved numerical value p_cap=1.4664687420450921; it is not moved to a different continuous Nash price when timing changes. The original 15-quote reference gives cap customer benefit approximately 72.119. The unchanged relaxed 95% target F is approximately 68.513, stored as an absolute numerical threshold. Different experiments do not quietly redefine this target.

Let q be the rival's previous completed price. The guard permits every price in the bounded domain whose immediate expected profit is within epsilon of the best immediate profit available against q:

```
S(q) = {p : max_z R(z,q) - R(p,q) <= epsilon + tau}
U(q) = max S(q)
```

Epsilon=0.1 in the base mask; the learner sensitivity uses epsilon=0.05. Tau is the inherited 10^-12 numerical comparison allowance. This is a lagged one-step modeled profit-loss rule, not calibrated regret and not loss against the rival's hidden current action. Both sellers are externally constrained; they may coordinate, randomize or use arbitrary history within the legal sets when we state an *all-policy* guarantee.

Table 1. Three evidence types and their distinct scopes.

Object	What is computed	What it does not establish
Saved finite game	Exact rational comparisons on the saved binary64 payoff weights and legal masks.	Exact-real logit inequalities or grid-invariant protection.
Validated continuous model	Directed Decimal interval bounds for imported real constants under checked monotonicity and contraction conditions.	An independent transcendental backend or empirical market validity.
Learning behavior	Fixed-budget local Q-learning, paired seeds, saved-policy continuation and visit support.	Convergence, equilibrium, intent, or LLM behavior.

In the graphical cap game a saved 61-quote base mesh is augmented by the exact old cap, prioritizing that quote over a nearby mesh quote. The two learner masks each have 62 choices. These are separate numerical games from the original 15-quote and refinement meshes; a continuous period is independently checked in the finite game before any finite guarantee is asserted.

Finite masks inherit the binary64-rounded epsilon+tau comparator. Continuous validation separately treats imported constants as exact reals and reports nominal/effective allowances. “Continuous limit” describes the model result, not a proof that bit-level JavaScript mask rounding vanishes.

The three outcome levels

An all-policy floor concerns every allowed path. A training average concerns the actual exploratory updates. A frozen continuation stops updates and exploration and follows the saved greedy choices. These answer different questions, and the paper and game keep them visible separately.

3 The handover promise weakens with richer prices

Under the verified base continuous assumptions, U is monotone and contractive on the inherited interval. It has a unique fixed point q*. Prices on every permitted path are bounded above by iterating U, and the upper choices themselves are permitted. Since customer surplus decreases in either price, C*=C(q*,q*) is a tight worst long-run floor. This does not require every permitted price path to converge.

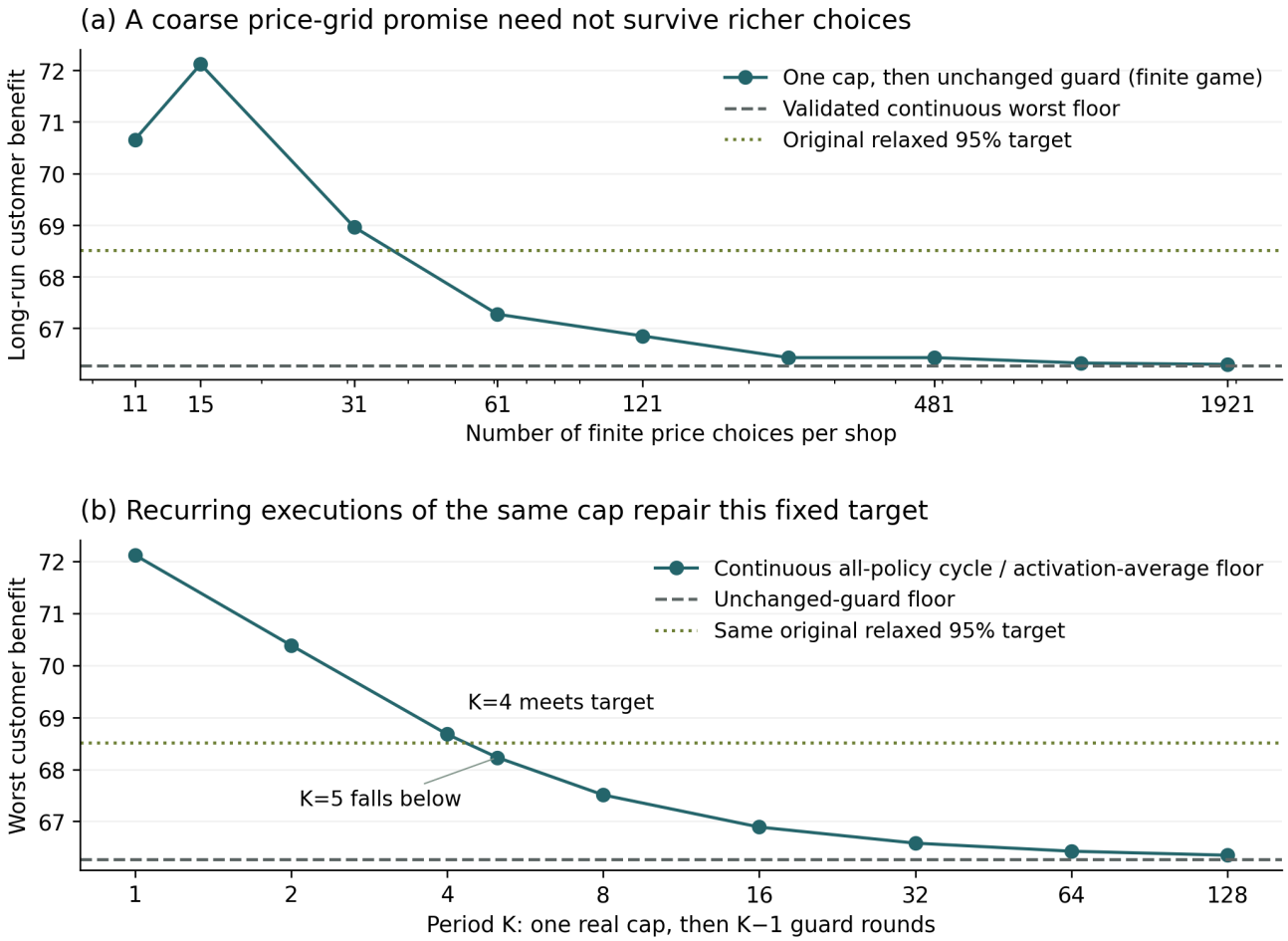


Figure 1. Base condition. (a) Saved finite one-cap floors at the predeclared mesh sizes and the validated continuous floor. Refinement is not monotone across all coarse meshes; the 11-choice case is already worse than the 15-choice case. (b) A different object: validated continuous recurring-cap floors G_K, with the same old physical cap and unchanged target. These plots show all-policy capacity, not observed learning.

The 15-choice one-cap floor is about 72.119, while the 1,921-choice diagnostic is about 66.298 and the continuous limit is 66.272. The original target is 68.513. A finite initializer changes a transient; it cannot raise the asymptotic continuous floor above C* when the unchanged guard then runs forever.

A tight repair within the fixed-period family

Keep physical price units: let a_0=p_cap and a_(j+1)=U(a_j), and put r_j=C(a_j,a_j). The actual first cap is phase 0; K-1 guard rounds follow. The worst repeated cycle is attained by both shops choosing their maximum legal quote:

```
G_K = (r_0 + ... + r_(K-1)) / K
```

The verified sequence r_j decreases, so every cap-first finite running average is at least G_K; a later individual round can still fall below it. The base target permits K=4 (G_4 about 68.676), but K=5 fails (about 68.229). Both 99% and 100% reference targets need K=1 in these audited labels. “Largest safe K” means largest *fixed integer period in this same-cap family*. Frequency 1/K counts rule applications, not economic cost, changed quotes or autonomy.

4 Two boundaries of the protection claim

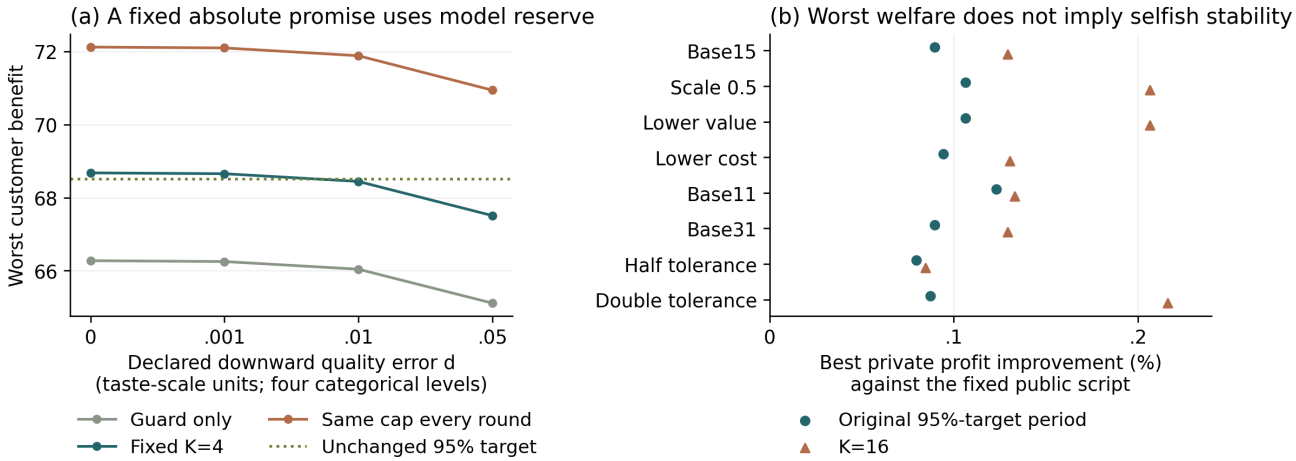


Figure 2. Separate sensitivity and incentive analyses. (a) The controller and old absolute target remain nominal while actual inside-good quality is v-d*mu. Four declared d values are shown categorically. Quality loss itself reduces customer benefit. (b) Exact best private-profit improvements against a fixed public maximum-allowed-price opponent, seller A, discount 19/20, across eight saved condition labels. These labels are not independent sampled economies; dots are not learner trials.

A fixed promise consumes model reserve

To isolate one explicit modeling assumption, actual demand quality is lowered by d taste-scale units while the legal rule, price domain, cap, clock and old target remain unchanged. The same price envelope remains the worst permitted path. If d is an unknown fixed value in [0,dbar], monotonicity makes dbar the worst endpoint; this is a bounded structural sensitivity calculation, not adversarial seller control over demand.

For the base original-95% rhythm K=4, the positive quality-error reserve is approximately d=0.006963. At dbar=0.01, fixed K=4 has worst customer benefit about 68.442, below the unchanged 68.513 target. Redesigning timing conservatively within the same family gives K=3 at dbar=0.01 and K=2 at dbar=0.05; at zero error the selected K is 4. Permanent use of the same cap retains more reserve. Tiny 100%-target radii mostly reflect the inherited numerical target margin, and should not be sold as a surprising economic mechanism.

A worst permitted path is not automatically selfishly stable

Against an opponent that publicly chooses U of the focal shop's previous price, the focal seller can optimize all its own permitted quotes over a cap-to-cap block. Exact finite dynamic programming checks every state and action, with a continuation through the next real cap. Seller B uses the correctly transposed physical rewards. The fixed opponent's cap choice makes block restarts history-independent; that special simplification does not transfer to learned cap choices.

In the base K=4 game, seller A earns 86.438659 by copying the script versus 86.516111 from its best allowed discounted block. The relative improvement is only 0.0896%. Across all eight labels, every recurring seller/period comparison at discount 19/20 has a positive but small activation gain; permanent-cap comparisons are mostly zero. A simple last-guard-round immediate deviation explains every recurring-script instability found by the full calculation; a separate immediate cap screen covers the exceptional K=1 case.

A positive gain rejects that specified fixed-script profile, not the controller's customer-benefit guarantee. A zero gain is not an equilibrium proof. Closely related unilateral pricing-policy audits already exist [4]; our BigInt dynamic program is an exact implementation for this finite rule, not a new algorithm.

5 What independently adapting local learners do

The previous results quantify what is allowed, not what a learner will discover. We therefore froze a prospective study before training on either study game: two saved 62-price masks in the same economic calibration, four schedules, two Q initializations and twelve fresh paired whole-market seeds. All 192 runs were kept, with no outcome-driven horizon or seed changes.

Table 2. Prospectively frozen adaptive-learning design; 192 retained runs.

Factor	Predeclared values
Guard masks	Original epsilon 0.1; half tolerance epsilon 0.05. Same economic calibration.
Schedules	One startup cap then guard forever; cap every round; inherited 95%-target rhythm (K=4 or 10); K=16. All begin with an actual cap.
Private scores	Zero; optimistic value 20 times that firm's largest own immediate profit.
Seeds and budget	907001–907012, paired actor streams across conditions; 2,000,000 simultaneous rounds per run.
Learning parameters	Alpha 0.1, gamma 19/20 represented as binary64; exploration declines exponentially from 1 to 0.01, without cap/checkpoint reset.
Saved endpoints	1M secondary and 2M primary checkpoints; exact last 100k training windows; complete frozen-policy graphs; 128-step terminal previews.

Each firm stores a private Q table on (cap/guard phase, own previous price, rival previous price). Both choose before either current action is revealed. Each observes only its own deterministic expected profit for updating; no teacher, shared reward or communication scripts the choices. Current greedy selection respects the current legal mask; the bootstrap maximum uses the next state and its actual next-clock mask. Exact ties select the smallest original quote index.

```
Q(s,a) <- Q(s,a) + 0.1*[R_own + 0.95*max_(legal a_next) Q(s_next,a_next) - Q(s,a)]
```

A cap is continuing, not terminal, and the state still retains price history at cap rounds. Thus a learned cap action can depend on incoming history. Counts are saved exactly, including saturation overflow records. Two private xoshiro128** streams per seed are prospectively separated by 2^64 jump segments, and paired across conditions. Ordinary binary64 arithmetic is used for learning; there is no rational-learning claim.

The primary comparison does not secretly add a new cap

For each tolerance and initialization, compare the final frozen-greedy eventual-cycle customer benefit of the selected rhythm with startup-only, using the *actual saved terminal history and phase*. Startup-only remains in guard forever. All twelve paired differences remain in each of four strata. A new high-history, phase-0 cap start is a separately labeled counterfactual, never the primary endpoint.

The prospectively saved 5,120 by 12 whole-seed resample matrix gives approximate marginal percentile 95% seed-variation intervals, using type-7 interpolation. These are not simultaneous intervals, market uncertainty, convergence evidence or a causal effect of frequency alone. Increasing K also increases phase-dependent Q rows; equal rounds do not mean equal state/action coverage. There is no unrestricted learner-capability arm, so this design cannot show that the guard prevented learned collusion.

6 All four planned learner comparisons

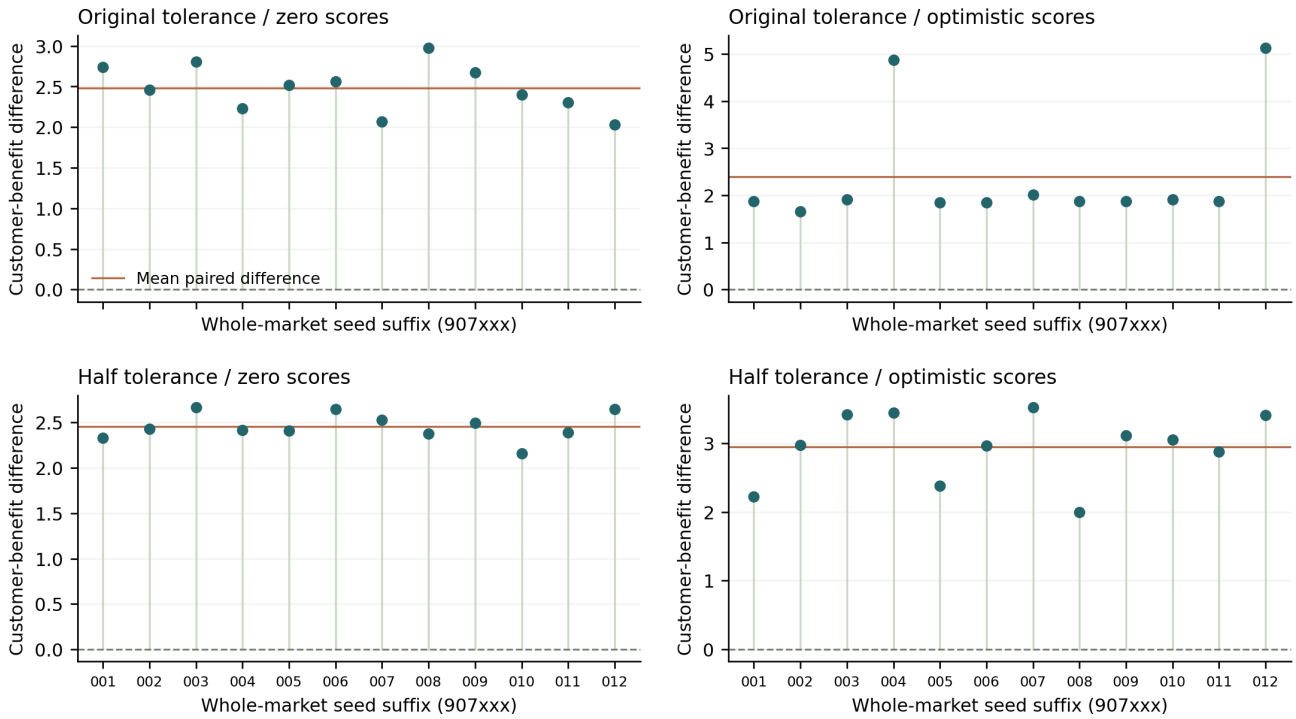


Figure 3. Every final 2M paired frozen-cycle customer-benefit difference: inherited selected95 rhythm minus one startup cap. Twelve whole-market seeds per panel; zero and reversed outcomes are retained. The horizontal line is the mean, not a protection floor. Tolerances and initializations remain separate; no seed, firm, state or checkpoint is counted as an additional independent economy.

Table 3. Final actual-terminal frozen-cycle differences, selected95 minus startup-only; all twelve paired seeds in each stratum.

Tolerance	Scores	Mean difference	All-seed range	Marginal 95% interval
Original	zero	+2.480	+2.031 to +2.978	+2.317 to +2.638
Original	optimistic	+2.391	+1.663 to +5.135	+1.850 to +3.160
Half	zero	+2.457	+2.158 to +2.664	+2.377 to +2.536
Half	optimistic	+2.946	+1.998 to +3.517	+2.657 to +3.197

Across the 48 planned paired rows, 48 differences are positive, 0 exactly zero and 0 negative. These four-stratum totals are descriptive bookkeeping, not a pooled effect estimate. Exact signs come from saved-weight fractions; intervals and printed decimals are approximate. Full seed rows remain in the evidence file.

The actual last 100k exploratory windows also favor the selected rhythm in 48/48 pairs. Four secondary mean gaps, in Table 3 order, are +2.321, +1.979, +2.446, +3.018. These are observed training-window averages, separate from frozen cycles and the primary interval calculation.

At the final checkpoint, 0/192 actual cycles contain an unvisited state and 0/192 a never-tried greedy action. Yet 152/192 change cycle customer benefit between 1M and 2M, and 21/192 differ under fresh-cap versus actual-terminal starts. These secondary diagnostics are not independent samples or convergence evidence. The display retains exploratory windows, both profits/prices, mask geometry, visit support and separate model floors.

All 192 runs were retained: 384 million market rounds, 768 million private updates and 12,288 discarded clone steps. Execution took 331.57 seconds; the largest after-run RSS sample (407,805,952 bytes) exceeded the 256 MB soft target. It is not a measured instantaneous peak. Full resource records remain in completion.json.

Independent Python reconstructs all 384 saved-Q greedy policy pairs, complete successor graphs and exact saved-weight cycle metrics, checks actual terminal envelopes, counts and diagnostic cloned updates, and rejects corrupted evidence. The numerical core was tested before training; its full schema adapter was completed and snapshotted during training before this independent referee read its first real checkpoint. The complete main training trajectory was not independently replayed; sampled updates and cloned probes were checked. No new learner economic outcomes were interpreted before the complete learner audit passed.

7 What this adds, and what remains open

The strongest result is about *the scope of a protection promise*. A coarse finite game can give a reassuring one-cap result while a richer price interface admits worse allowed continuations. The continuous upper envelope explains the limit and gives an attainable worst cycle when the same physical cap is repeated. Calibration reserve and unilateral incentives identify different boundaries: model error can consume the old absolute target, while a worst permitted script need not be individually attractive.

The learner study adds behavior under a precisely declared controller, without replacing the all-policy proof by a favorable simulation. Its outcomes remain conditional on two masks, one calibration, chosen hyperparameters, finite budget and private-score initializations. Neither high prices nor a quiet frozen cycle establishes collusion, punishment, intent or equilibrium. A future experiment needs an unrestricted capability reference, systematic learning sensitivity and an explicit stability audit against the actual learned opponent before making those claims.

Relation to prior work and the sprint starter

Calvano et al. study profit-seeking learning and algorithmic pricing [2]. Eschenbaum et al. examine robustness, replacement and continued adaptation [3], and the sprint starter already branches histories, models and objectives [1]. Alshiekh et al. define action restriction and shielding, including reward credit and the need for persistent protection [5]. Brero et al. design ongoing platform interventions for consumer benefit [6]. Hartline et al. study a different, calibrated-regret regulation framework [7]. Periodic resetting and renewal-ratio control are established [8,9]. Lopes de Moraes and De Losso directly audit learned pricing incentives [4]. None should be repackaged here as our invention.

Our limited addition is a linked, reproducible guarantee-capacity case: price precision, a fixed physical cap, all-permitted-policy timing, bounded demand reserve and adaptive observations are evaluated with their distinct quantifiers. We do not establish broad novelty, empirical prevalence, superiority to prior platform mechanisms or a globally optimal intervention. The largest fixed integer period is only a family result; state-dependent or mixed-gap schedules remain a separate research question.

An accessible game, with evidence behind each view

Start at http://127.0.0.1:3003/. The two-shop protection challenge lets a human coordinate all allowed quotes, see actual cap rounds and inspect the running customer-benefit buffer. The one-shop profit challenge uses a transparent public opponent and exact one-block score. The learner-evidence page reveals stored frozen-policy executions with both shops, purchases and profits; it never trains a player in the browser. These are explanatory artifacts and developer-tested controls, not a human usability study.

Reproduction

From the project folder, start the read-only local viewer with `node research/demo-server.js`. The README identifies the frozen protocols, raw manifests and independent checkers. Run `python research/audit-adaptive-learning.py` to check saved learner artifacts; run the documented earlier checkers for the continuous, recurring, calibration and exact-incentive claims. Rebuild derived learner summaries and graphics only after current audit gates pass. The portable archive preserves inherited checkpoints rather than silently replacing them.

Verification checks are coverage evidence, not experimental sample size. The exact finite calculations share their economic model; the continuous checkers use separate directed-interval code but the same Decimal transcendental backend. Independent AI-agent review does not substitute for external human peer review.

References

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Event and document context

Prepared for the AI Collusion Research Sprint organized by Apart Research. The public sprint page and template guide the question and report structure; this local PDF has not been submitted, externally deployed or endorsed. Affiliation has not been supplied by the named author and no institution is invented.

Official sprint page · Public report template

Appendix: Limitations and Dual-Use / Ethical Considerations

Model limits. The mathematical statements require the explicitly bounded, symmetric, known logit model and checked upper-map assumptions. The saved finite results certify numerical masks and reward weights, not exact-real profit inequalities. The continuous results concern imported constants and directed intervals, not measured economic uncertainty. The quality-error test is one fixed structural perturbation, not general robustness. Frequency counts external rule applications and does not measure implementation cost or useful seller agency.

Behavioral limits. The new 192-run study has one economic calibration, two tolerance masks, twelve paired seeds and fixed learning settings. Changing phase count changes state space and coverage. Deterministic expected rewards omit sampled demand. Frozen eventual cycles may contain default unvisited scores. No unrestricted capability arm, policy-punishment mechanism test against learned rivals, convergence proof, equilibrium proof, LLM comparison or human participant study is included. A positive fixed-script unilateral gain and a customer-benefit floor answer different questions.

Verification limits. Separate implementations reconstruct saved finite policy/graph/incentive calculations; they can share a conceptual or modeling error. Continuous primary and independent code share a Decimal transcendental backend. Learner source, checkpoints, counts and cloned probes were audited, but the complete main learning trajectory was not independently replayed; sampled updates and cloned probes were checked. Sampled after-run RSS exceeded the prospective soft memory target and is not a measured instantaneous peak. All declared runs and resource overruns remain visible.

Dual use and ethics. The toy model can illustrate adverse price coordination, so outputs should be framed as tests of safeguards rather than operational coordination advice for real firms. No real seller communication, customer data, participant recruitment, private economic data or autonomous commercial trading is involved. Human controls operate locally without claiming experimental observations. No external publication, submission, paid inference or deployment was performed in this stage.

LLM usage and author contributions

The project builds on the user's preexisting browser game and market scaffold, and on published sprint resources. Those are prior work. The guarantee-capacity analyses, new controller demonstrations and 192-run learner study described here were developed in the current sprint work; frozen protocols, source snapshots and timestamps identify that new work. The report does not claim the inherited game or published starter as a new contribution.

Mohammadpayam Teabi supplied the project, direction and author identity. Codex and collaborating AI agents substantially performed problem formulation, code implementation, mathematical derivation, experiment orchestration, independent AI-agent checks, analysis, visualization and drafting. The report is substantially AI-written; independent AI agents are not independent human peer reviewers. No human validation of the final science is implied.

The new guarantee-capacity analyses and 192-run tabular experiment use no experimental language-model seller, paid model service, account-backed game player or remote inference. Codex development consumes coding-assistant allowance. Historical account-backed pilot material exists elsewhere in the earlier project and is not evidence for the claims in this report; the report does not claim that the entire project history had zero model usage.

Compact provenance pointers

Learner frozen manifest: research/adaptive-learning-results/manifest.json
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Learner independent receipt: research/adaptive-learning-results/independent-audit.json
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